

CHE 2201 – Physical Chemistry II

Phase Equilibria

Lec. 6. Solid-Liquid Equilibria

6.1. Introduction

Under the binary systems, we have studied the mixtures of two liquids, i.e. liquid solutions of two components. Then, we have studied the liquid-vapour systems, which are in equilibrium with each other. Now, we intend to study the behaviour of solid-liquid systems, which are in equilibrium with each other.

Composition of a solid-liquid system plays an important role in deciding the transition between liquid and solid phases. The behaviour is very similar to azeotrope in liquid-vapour transition. The phase diagrams of solid-liquid systems at const. pressure indicates the significant changes of such systems.

6.2. Phase Diagram of a Solid-Liquid System – Simple Eutectic Diagram

If a liquid solution of two substances, A and B, is cooled at a certain temperature, a solid will appear. This temperature is the freezing point of the solution, and it depends on composition. We have already learnt that the addition of one component to another reduces the freezing point of the pure component.

If we draw a graph of freezing point temperature versus mole fraction (X_A or X_B), it will take the form of the diagram shown in Fig. 6.1.

The points above the curve represent liquid state of the system, and below the curve represent the state at which the pure solid A is in equilibrium with the solution. This curve is called the liquidus curve.

However, this curve cannot be considered as the entire freezing curve of the solution, since it contains the other component B. As X_B reaches 1, the component B also starts to freeze at this range of temperature, giving rise to a curve similar to that of A (indicated by

the dotted line in Fig. 6.1). So, the resultant curve of these two liquids and phases is given in Fig. 6.2.

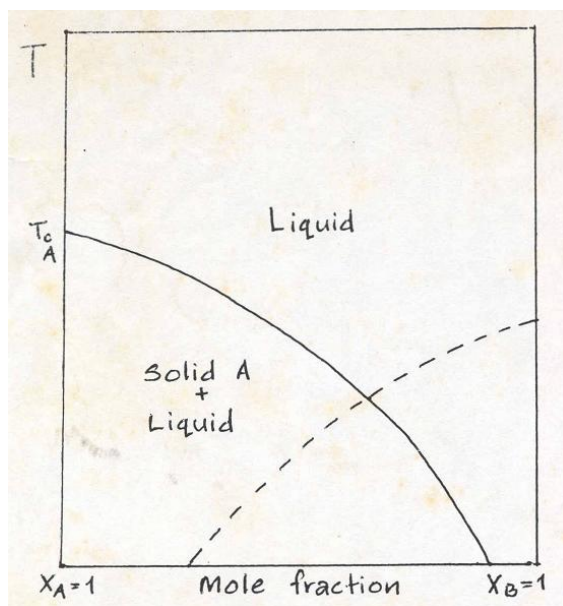


Fig. 6.1. Solid-Liquid Equilibria in a Two Component System

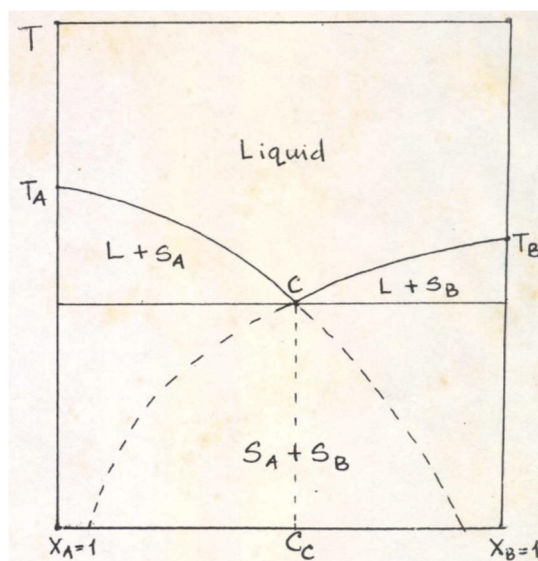


Fig. 6.2. Phase Diagram of a Solid-Liquid System

Figure 6.2 shows a phase diagram of a mixture with components A and B, which does not form a solid solution. On cooling, components A and B separate as pure solids.

The freezing points of the pure components are given as T_A and T_B . Curves T_AC and T_BC are called freezing point curves of A and B. Curve T_AC represents the change in the freezing point of A with respect to the different compositions of A and B. Similarly, curve

T_{BC} represents the change in the freezing point of B with the composition of the mixture. As such, the curves T_{AC} and T_{BC} tell us the equilibrium states of solid A and solid B with the solution.

At point C, both solids A and B are in equilibrium with a liquid solution of definite composition given by the point C_C . In other words, the point C indicates the conditions of temperature and the composition of the liquid phase, under which the system of solid A, solid B and liquid can exist in equilibrium (i.e. $S_A - S_B - L$).

This is the lowest temperature at which any mixture of solid A and B will melt. For this reason, the point C is called the **eutectic point**. (Eutectic is a Greek word which means “easily melts”). This point lies at a T, lower than the melting points of both components. At the eutectic point, the crystals of both components form simultaneously. At all temperatures above the curves T_{AC} and T_{BC} , the system exists as a homogeneous liquid system. At all temperatures below the eutectic temperature, only solid phases exist in equilibrium.

At eutectic point C:

Number of Phases (P) = 3;

Number of Components (C) = 2.

So, from the Phase Rule:

$$F = C - P + 2 = 2 - 3 + 2 = 1.$$

$\Rightarrow$ The system is univariant.

Here, the only variable is pressure, as at this point the composition and temperature are constants.

6.2.1. Example – Antimony–Lead (Sb–Pb) System

The antimony-lead system has the single eutectic type phase diagram, what we have discussed in the previous section.

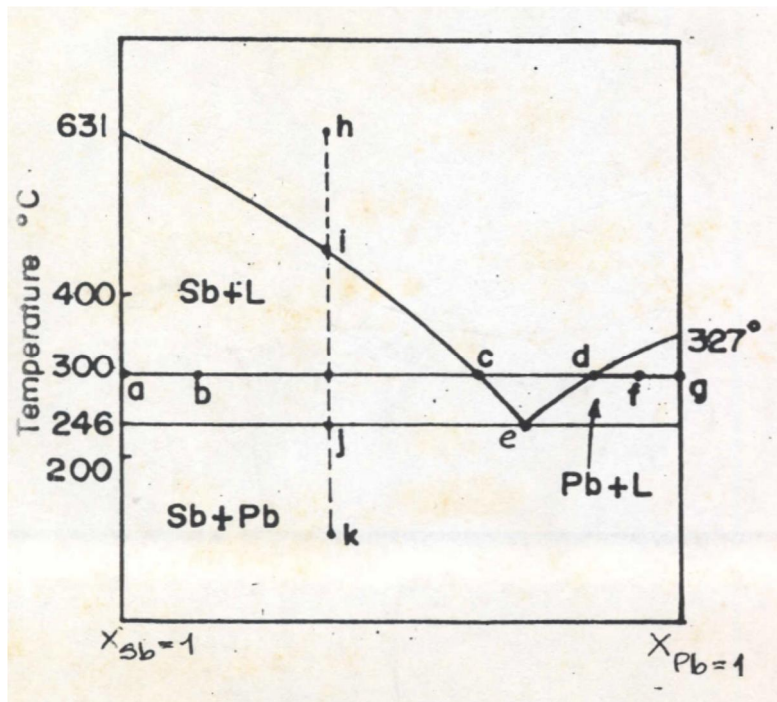


Fig. 6.3. Phase Diagram of Sb-Pb System

(a). Isothermal Behaviour

Let us consider the isothermal behaviour of the system at 300°C , which is given by the line $abcdfg$. The point 'a' represents the pure solid antimony at 300°C . The addition of Pb to Sb, forms a mixture of solid Sb that coexists in equilibrium with the liquid of composition c. All the added Pb melts and the molten Pb dissolves enough of the solid Sb to bring the liquid to composition c. Therefore, at point 'b', the relative amount of liquid present is quite small, and is in equilibrium with the solid.

Addition of more Pb dissolves more of solid Sb to form the solution of composition c, and the point moves from b to c. At point 'c', sufficient amount of Pb has been added to dissolve all the original Sb present at a, and the final solution is a saturated solution of Sb in Pb. Addition of Pb beyond the composition c simply dilutes the solution at c, and the point moves through the liquid region from c to d, since the T is constant. At point 'd', the solution is saturated with Pb. At point 'f', the added Sb (let's assume that pure Pb is at 'g', and Sb is added to Pb), dissolves sufficient Pb to form a solution of d and excess Pb exists as solid Pb.

(b). Cooling Behaviour

Let us consider the cooling behaviour of a constant composition mixture, which is given by the line *hijk*.

At point 'h', the system is entirely liquid. As the system cools, solid Sb appears at point 'i'. Therefore, the solution becomes richer in Pb. As the T moves from i to j, the liquid composition moves along the curve from i to e. At point 'j', the solution has the eutectic composition and is saturated with respect to Pb. Hence, Pb begins to precipitate. At the eutectic composition, the T remains constant until the liquid solidifies. Finally, the T of two solids decreases along the line *jk*.

From the above discussion, it is clear that the eutectic composition has the lowest melting point. The eutectic mixture melts sharply at the eutectic T to form a liquid of the same composition, whereas other mixtures melt over a range of Ts. It has been observed that eutectic mixture is a homogeneous mixture and not a compound. This technique is used in metallurgy to separate pure solid from solid mixtures.

6.3. Thermal Analysis

The phase diagrams of binary systems can be constructed experimentally by thermal analysis. In this method, a mixture of known composition is heated to a higher T until it melts to a homogeneous liquid. It is then allowed to cool steadily, and the T is recorded at regular intervals of time, so as to obtain the cooling curve (Fig. 6.4).

From point 'a' to 'b', the homogeneous liquid cools rapidly. Then at point 'b', solid A **commences** to separate, and this releases the heat of fusion. Therefore, the rate of cooling is slowed down; the cooling curve thus shows a break at temperature 'b', i.e. the freezing point where solid first commences to form. As the composition of the liquid phase changes, the T falls from 'b' to 'c'. At point 'c', the liquid has the eutectic composition and solid B starts to form. At this point, T remains constant until all the liquid solidifies at point 'd'. The horizontal line *cd* indicates the constant T known as the eutectic halt.

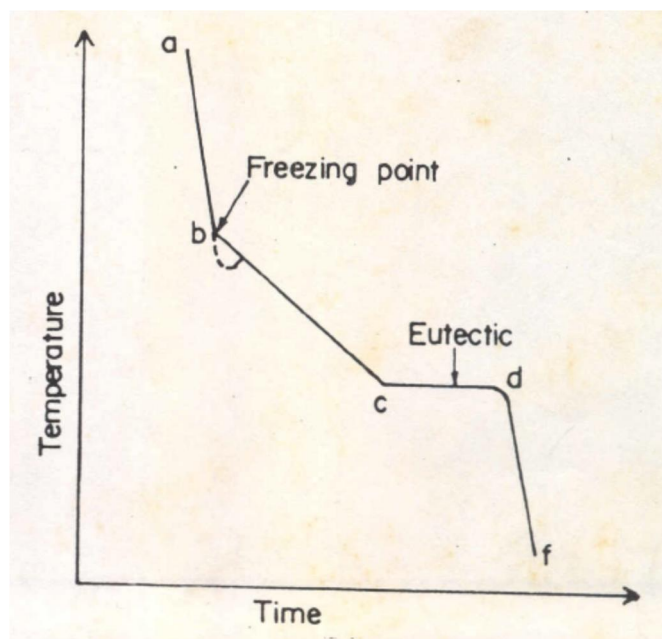


Fig. 6.4. Cooling Curve

Frequently, due to the absence of crystallization nuclei, supercooling occurs and the T drops slightly below 'b' without separating. But, as soon as solid begins to form, heat is evolved and there is a transient increase of T, as shown by the dotted line in Fig. 6.4.

From the cooling curve for a mixture of any definite composition, it is possible to obtain the freezing point of that mixture and the eutectic T. Having obtained the cooling curves for different compositions, we can construct the phase diagram of the system (See Fig. 6.5).

The first two curves are for liquid mixtures richer in A than B. Here, the interpretation is same as above, except the eutectic halt and the freezing T. The third curve illustrates the cooling of the eutectic mixture. Here, the region bc will not exist; the cooling curve will not show a break until the eutectic T is reached. In this case, the eutectic halt has its maximum length.

The 4th, 5th and 6th curves are for the compositions richer in B than A. T_4 , T_5 and T_6 are the temperatures, where first crystals of B are formed. Again the length of the eutectic halt decreases as the composition departs from the eutectic composition. To construct the phase diagram, the temperatures T_1 , T_2 , T_4 , T_5 , T_6 and T_e (eutectic temperature) are plotted against the composition.

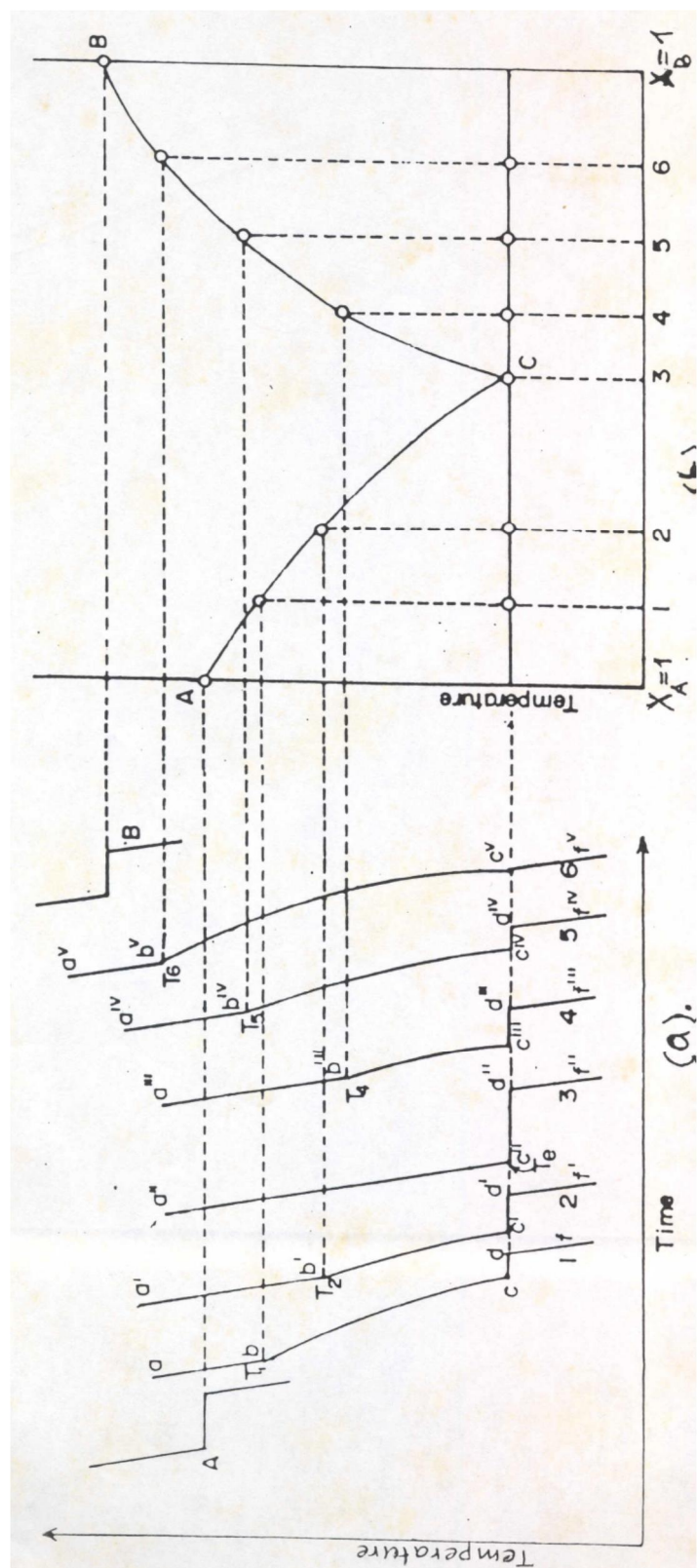


Fig. 6.5. Construction of a Phase Diagram by Thermal Analysis
 (a). Cooling Curves of Pure A and B and Mixtures of Compositions 1 – 6;
 (b). Phase Diagram

6.4. Phase Diagrams with Compound Formation

In previous sections, we have discussed two component systems which do not form compounds during the cooling or melting of the system. However, in some cases, the components of binary systems form chemical compounds having definite melting point. These compounds are frequently stable and melt without decomposition. In other words, the solid compound is capable of existing in equilibrium with a liquid of the same composition. Such melting is called congruent. (Congruent means exactly the same). So, the congruent melting point is the point at which the solid compound is in equilibrium with liquid of the same composition.

6.4.1. Compound Formation with a Congruent Melting Point (Without Decomposition)

If two substances form one or more compounds, the phase diagram has the appearance of two or more simple eutectic diagrams. Fig. 6.6 represents the freezing point diagram (phase diagram) of a system in which a compound AB_2 is formed.

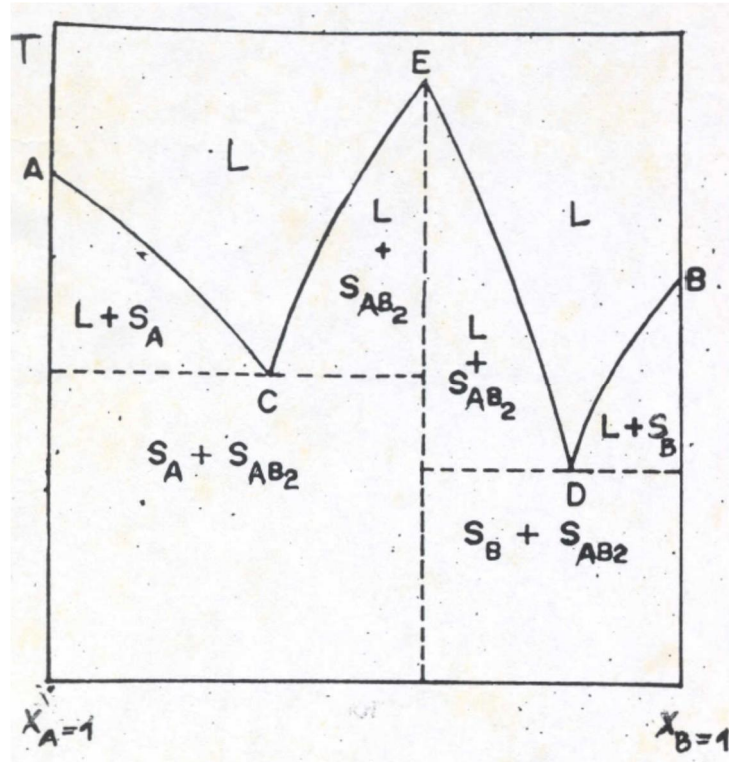


Fig. 6.6. Compound Formation with a Congruent Melting Point

Points A, E and B are congruent melting points of component A, compound AB_2 and component B respectively. Curve AC indicates the variation of composition of the liquid mixture, which is in equilibrium with the solid component A, with temperature. Curve BD indicates the variation of composition mixture, which is in equilibrium with solid component B, with temperature.

CED – Composition of solutions in equilibrium with the solid component AB_2 ;

C – Eutectic point of components A and AB_2 ;

D – Eutectic point of components B and AB_2 .

The melting point of compound AB_2 must be a maximum point on the equilibrium curve CED. This may be either higher or lower the melting point of either of components, or it may have an intermediate position. If the compound AB_2 is stable, or the degree of dissociation is less in the liquid phase, then the bend of the curve (at point E) will be sharp. If the degree of dissociation is high, then the curve appears less sharp at point E.

If several compounds are formed with congruent melting points, then a series of curves similar to CED will appear in the phase diagram. The final phase diagram will be a combination of these simple eutectic phase diagrams. For example, $FeCl_3$ forms several hydrates of compositions $FeCl_3 \cdot 6H_2O$, $FeCl_3 \cdot 3\frac{1}{2}H_2O$, $FeCl_3 \cdot 2\frac{1}{2}H_2O$ and $FeCl_3 \cdot 2H_2O$. Hence, the final phase diagram of this system is a combination of five simple eutectic diagrams as shown in the Fig. 6.7.

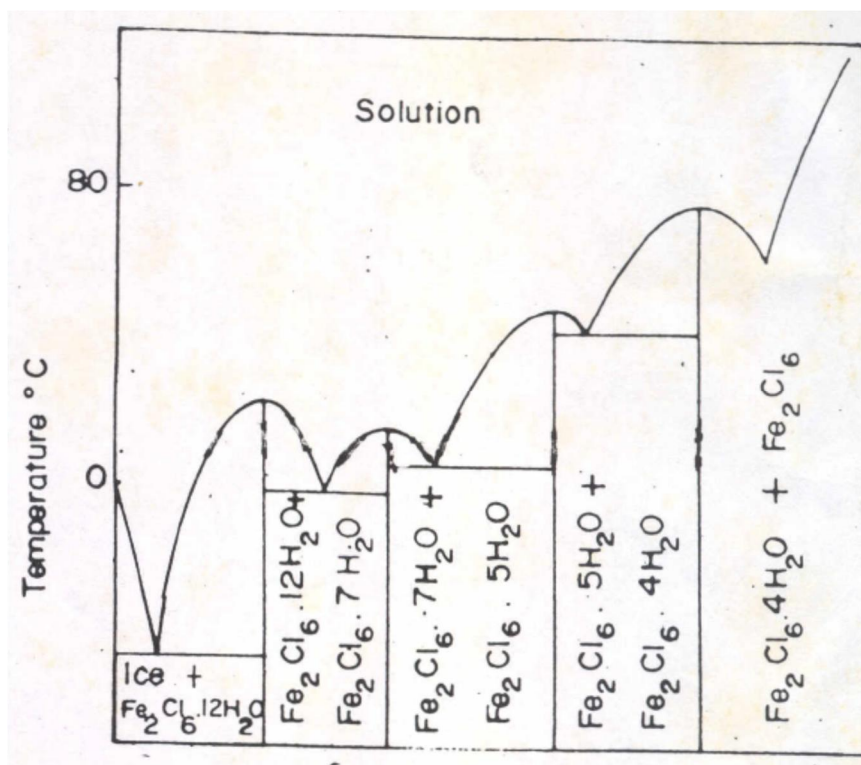


Fig. 6.7. Schematic Phase Diagram of the System $\text{FeCl}_3\text{-H}_2\text{O}$

6.4.2. Compound Formation with an Incongruent Melting Point (With Decomposition)

In the previous section, we have discussed the system which forms a compound of higher melting point than both components. However, if the melting point of the compound lies below the melting point of one of the components, two possibilities arise. The first is illustrated in the figure 6.7 (system $\text{FeCl}_3\text{-H}_2\text{O}$), where each part of the diagram is a simple eutectic diagram.

The second possibility is the decomposition of the compound into solid and liquid phases of different compositions. Such a melting is called incongruent, i.e. melting in which the compositions of initial solid phase and the composition of the formed liquid phase differ.

One such example is the mixture of Na and K, and the phase diagram of this mixture is shown in Fig. 6.8.

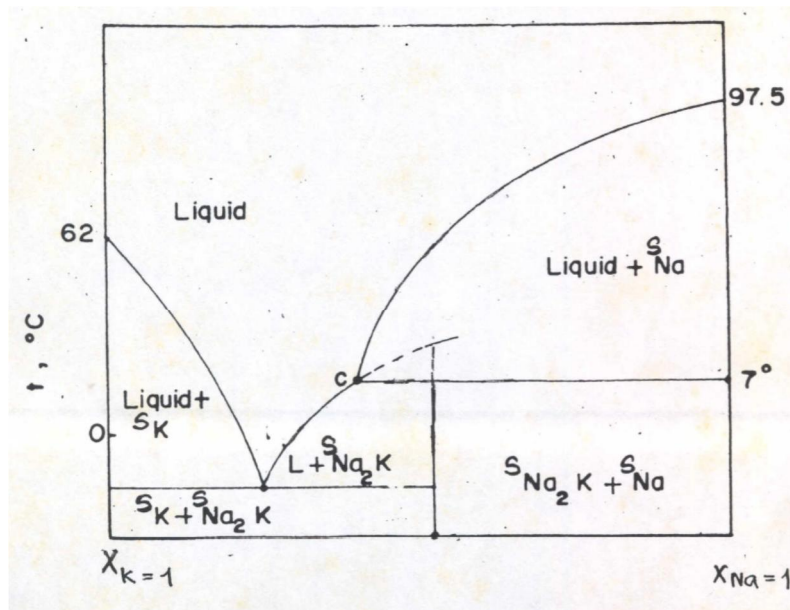


Fig. 6.8. Compound Formation with an Incongruent Melting Point – Na-K System

The peak of the dotted curve is the expected congruent melting point of Na_2K . But, before reaching this point, the solid Na_2K decomposes and changes into another solid phase and a liquid at C. The composition of the liquid formed is not the same as that of the compound. The point C is called the peritectic point. It represents the maximum T, at which solid Na_2K can exist under that particular pressure.